

SECTION A

SECTION B

SECTION C

1.
 - a) What do you understand by stable and in place shorting?
 - b) How does bubble short work? Explain.
 - c) Differentiate sequential search and binary search.
 - d) Give example of one each stable and unstable sorting techniques.
 - e) Illustrate the concept of Indexed Sequential search.
- SECTION B**
2.
 - a) Use quick sort algorithm to sort 15,22,30,10,15,64,1,3,9,2. Is it a stable sorting algorithm? – Justify.
 - b) How binary search is different from linear search? Apply binary search to find item 40 in the sorted array: 11,22, 30, 33,40, 44,55, 60, 66,77, 80, 88, 99. Also discuss the complexity of binary search.
 - c) Explain any three commonly used hash function with the suitable example. The highest function H defined as $H(\text{key}) = \text{key} \bmod 7$, with linear probing, is used to insert the key 37,38, 72, 48, 98, 11, 66, into a table indexed from 0 to 6. What will be the location of key 11? Justify your answer, also count the total number of collisions in this probing.
 - d) Write algorithms of insertion sort. Implement the same on the following numbers; also calculate its time complexity. 13, 16, 10, 11, 4, 12, 6, 7
 - e) Differentiate between liner and binary search algorithm. Write a recursive function to implement binary search.
 - f) Write an algorithm for Quick sort. Use Quick sort algorithm to sort the following elements: 2, 8, 7, 1, 3, 5, 6,4.

- 3.** (a) Write short notes on:
- (b) Explain the following:
- I. Heap Sort
- II. Radix Sort.
- Hashing Technique

- 4.(a) Write algorithm for quick sort. Trace your algorithm on the following data to sort the list: 2,13,4,21,7, 56,51, 85,59, 1,9,10. How the choice of pivot element effects the efficiency of algorithm.
- 5.(a) Why is quick sort named as quick? Show the steps of quick sort on the following set of elements: 25,57, 48,37,12,92,86,33. Assume the first element of the list to be the pivot element.

- (b) What is hashing? Give the characteristics of hash function. Explain collision resolution technique in hashing.
6. (a) Write an algorithm for merge sort and apply on following elements. 45,32, 65,76, 23,12, 54, 67, 22. 87.
 (b) Write a C programme for index sequential search .
7. (a) (i) Use the merge sort algorithm to sort the following elements in ascending order. '13, 16, 10, 11, 4, 12, 6, 7. What is the time and space complexity of merge sort?
 (ii) Use quick sort algorithm to sort 15,22,30,10,15,64,1,3,9,2. Is it a stable sorting algorithm? Justify.
 (b) (i) The keys 12, 17, 13, 2, 5, 43, 5 and 15 are inserted into an initially empty hash table of length 15 using open addressing with hash function $h(k) = k \bmod 10$ and linear probing. What is the resultant hash table?
 (ii) Differentiate between linear and quadratic probing techniques.
- 8.(a) What is Hashing? Explain division method to compute the hash function and also explain the collision resolution strategies used in hashing.
 b) Write an algorithm for Heap Sort. Use Heap sort algorithm, sort the following sequence: 18, 25, 45, 34, 36, 51, 43, and 24.
- 9 a. Construct a Hash table that contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is $key \% 10$. Insert values 43, 165, 62, 123, 142 in the table. Show all the steps and find the location of the key value 142?
 b. Demonstrate merge sort algorithm to sort the following elements in ascending order : 11, 16, 13, 11, 4, 12, 6, 7. What is the time and space complexity of merge sort.